

to be sent between computers. It is speculated that e-mail had been around since 1965: a program was developed by Fernando Corbató and colleagues to swap messages amongst the individual users of the CTSS at MIT, but communication between machines was possible only with those terminals on which the System was installed.

(Later, on the request of the director of ARPA, Steve Lukasik, an e-mail management program called RD was written by Larry Roberts in July of 1972 as an improvement upon READMAIL. RD enabled the listing of incoming messages, and supported replying, forwarding, and filing them in any order.)

November: The Big Landmark—Intel’s 4004 Microprocessor

The CPU, which is often referred to as the brain of a computer, happens to be a microprocessor. Today, besides computers, thousands of products carry a microprocessor: consumer electronics, portable items, mobile phones, embedded systems, and more. The first microprocessors were designed as components for electronic calculators. The invention of the Integrated Circuit in the late 1950s had revolutionised the second generation of computers in the 1960s.

If you remember, a Japanese calculator manufacturer called Busicom had signed, in 1969, a contract with Intel to develop a dozen ICs for an electronic calculator—for their five calculating machines. A team of three engineers—Federico Faggin, Marcian E “Ted” Hoff, and Stanley Mazor—was put to the task. But instead of making a dozen chips, the team came with a chip design that included four chips: nine months of hard work resulted in the 4004, a 16-pin microprocessor that was one-eighth by one-sixteenth of an inch large. It had 2,300 metal-oxide transistors and was as powerful as the ENIAC (built in 1946).

The tiny four-chip design was packed with the 4004 (a central processing unit chip, the CPU); the 4001 (a read-only memory chip, the ROM); the 4002, a RAM chip; and a shift register chip for